lec-04

* Chain rule of Differential calculus:

If $y = f(u)$ and $u = g(x)$ then

$$\frac{dy}{du} = f'(u) \quad \text{and} \quad \frac{du}{dx} = g'(x).$$

and $$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Also, If $y = f(u)$; $u = g(v)$; $v = h(x)$

$$y \to u \to v \to x.$$

Then, $$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx}.$$

Example: Evaluate $\frac{dy}{dx}$ when

$$y = \ln(\underline{\cos(x^3)})$$

Ans: Given, $y = \ln(\underbrace{\cos(x^3)}_{u})$

$$y = \ln u$$

$$\Rightarrow \frac{dy}{du} = \frac{d}{du} \ln u = \frac{1}{u} = \frac{1}{\cos(x^3)}$$

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let ~~$y = f(u) = \ln$~~

$$u = \cos^{-1}(x^3) \quad ; \quad v = x^3$$

$$\Rightarrow u = \cos(v)$$

$$\Rightarrow \frac{du}{dv} = \frac{d}{dv}\cos(v)$$

$$\Rightarrow \frac{du}{dv} = -\sin(v)$$

$$\therefore \frac{du}{dv} = -\sin(x^3)$$

$$\Rightarrow \frac{dv}{dx} = \frac{d}{dx}x^3$$

$$\Rightarrow \frac{dv}{dx} = 3x^2$$

so, $$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx}$$

$$= \frac{1}{\cos(x^3)} \cdot \{-\sin(x^3)\} \cdot 3x^2$$

$$\therefore \frac{dy}{dx} = -3x^2 \tan(x^3)$$

Ans.

Ex. Evaluate $\frac{dy}{dx}$, where

$y = \tan^2 x$.

Ans. Given,

$y = (\tan x)^2$

We know the derivative of x^n where $n \in \mathbb{R}$. But there the power (square) is on $\tan x$ not on only x.

So, we will replace $\tan x$ by a new variable

i.e, let $\tan x = u$, then

$y = u^2$; $u = \tan x$

Now we can calculate the derivative of y with respect to u and the derivative of u

with respect to x.

Here, $y = u^2$ | $u = \tan x$

$\Rightarrow \frac{dy}{du} = 2u$ | $\Rightarrow \frac{du}{dx} = \sec^2 x$.

$\therefore \frac{dy}{du} = 2\tan x$

So, $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

$= 2 \cdot \tan x \cdot \sec^2 x$.

In short form

$$\frac{d}{dx}(\tan x)^2 = \frac{d(\tan x)^2}{d(\tan x)} \cdot \frac{d(\tan x)}{dx}$$

$$= 2\tan x \cdot \sec^2 x.$$

* $\frac{d}{dx}[\ln x] = \frac{1}{x}$

$$\frac{d}{dx}\left[\ln(x^2+1)\right] = \frac{1}{x^2+1} \cdot (2x+0) = \frac{2x}{x^2+1}$$

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<u>Example</u>: If $y = \left(1+\cos(x^2-1)\right)^{3/2}$ then evaluate $\frac{dy}{dx}$.

<u>Ans</u>:

$$\frac{dy}{dx} = \frac{d\left(1+\cos(x^2-1)\right)^{3/2}}{d\left(1+\cos(x^2-1)\right)} \cdot \frac{d\left(1+\cos(x^2-1)\right)}{d\,(x^2-1)} \cdot \frac{d(x^2-1)}{dx}$$

$$= \frac{3}{2}\left(1+\cos(x^2-1)\right)^{3/2-1} \cdot \left(0-\sin(x^2-1)\right) \cdot (2x-0)$$

$$= \frac{3}{2}\left(1+\cos(x^2-1)\right)^{1/2} \left(-\sin(x^2-1)\right)(2x)$$

$$\therefore \frac{dy}{dx} = -3x \cdot \sin(x^2-1)\left(1+\cos(x^2-1)\right)^{1/2}$$

<u>Ans.</u>

* ~~Explicit~~
* <u>Explicit function</u>:

An equation of the form $y = f(x)$ is said to define explicitly as a function of x because the variable y appears alone on

one side of the equation and does not appear at all on the other side.

$$y = \frac{x-1}{x+1}$$

$$y = \sin x \quad \text{etc.}$$

☆ Implicit function:

An implicit function is a function that depends on both x and y in such a way that the two variables cannot be separated across the equals sign without using more than one explicit function.

Ex. $5y^2 + \sin y = x^2$

☆ Implicit Differentiation:

$$5y^2 + \sin y = x^2$$

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$$\Rightarrow \frac{d}{dx}\left(5y^2 + \sin y\right) = \frac{d}{dx}\left(x^2\right)$$

$$\Rightarrow 5\frac{d}{dx}y^2 + \frac{d}{dx}\sin y = 2x$$

$$\Rightarrow 5 \cdot \frac{dy^2}{dy} \cdot \frac{dy}{dx} + \frac{d(\sin y)}{dy} \cdot \frac{dy}{dx} = 2x$$

$$\Rightarrow 5 \cdot 2y\frac{dy}{dx} + \cos y\frac{dy}{dx} = 2x$$

$$\Rightarrow \frac{dy}{dx}\left(10y + \cos y\right) = 2x$$

$$\therefore \frac{dy}{dx} = \frac{2x}{10y + \cos y}$$ Ans.

★ Leibnitz theorem:

Leibnitz rule is a generalization of the product rule. It states that if the functions $f(x)$ and $g(x)$ are differentiable n times then their product $f(x) \cdot g(x)$ is also differentiable n times.

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$$\frac{d^n}{dx^n}(fg) = \frac{d^n f}{dx^n}\cdot g + {}^nc_1 \frac{d^{n-1}f}{dx^{n-1}}\, g\, \frac{dg}{dx} +$$

$$ {}^nc_2 \frac{d^{n-2}f}{dx^{n-2}} \frac{d^2 g}{dx^2} + \cdots\cdots + f\frac{d^n g}{dx^n}.$$

$$= \sum_{K=0}^{n} {}^nc_K \frac{d^{n-K}f}{dx^{n-K}} \cdot \frac{d^K g}{dx^K}.$$

★

or, $u = u(x)$; $v = v(x)$; let $\frac{du}{dx} = u_1$, $\frac{dv}{dx} = v_1$

$\frac{d^2u}{dx^2} = u_2$, ... $\frac{d^n u}{dx^n} = u_n$. then,

$\cancel{\frac{d^n}{dx^n}(uv) =}$

$$\frac{d^n}{dx^n}(uv) = (uv)_n = u_n v + {}^nc_1 u_{n-1} v_1 + {}^nc_2 u_{n-2} v_2$$
$$+ {}^nc_3 u_{n-3} v_3 + \cdots\cdots + u v_n.$$

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★ **Example:** Find the nth derivative of e^{ax}.

Ans:

let $y = e^{ax}$

$$\Rightarrow \frac{dy}{dx} = y_1 = \frac{d}{dx} e^{ax}$$

$$\Rightarrow y_1 = a e^{ax}$$

$$\Rightarrow \frac{dy_1}{dx} = a \frac{d}{dx} e^{ax}$$

$$\Rightarrow y_2 = a \cdot a e^{ax} = a^2 e^{ax}$$

$$\Rightarrow y_3 = a^3 e^{ax}$$

$$\vdots$$

$$\Rightarrow y_n = a^n e^{ax} \quad ; n \in \mathbb{Z}^+$$

★ **Example:** Find the nth derivative of, $y = \sin(ax+b)$

Ans:

$$y = \sin(ax+b)$$

$$\Rightarrow y_1 = a \cos(ax+b)$$

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$$\Rightarrow y_1 = a\cos(ax+b)$$

$$\Rightarrow y_1 = a\sin\left(ax+b+\frac{\pi}{2}\right)$$

$$\Rightarrow y_2 = a^2\cos\left(ax+b+\frac{\pi}{2}\right) = \cancel{a^2\cos\left(ax+b+\frac{\pi}{2}+\frac{\pi}{2}\right)}$$

$$\Rightarrow y_2 = a^2\sin\left(\cancel{\text{or}}\ ax+b+\frac{\pi}{2}+\frac{\pi}{2}\right)$$

$$\Rightarrow y_3 = a^3\cos\left(ax+b+\frac{2\pi}{2}\right)$$

$$\Rightarrow y_3 = a^3\sin\left(ax+b+\frac{3\pi}{2}\right)$$

$$\vdots$$

$$\Rightarrow y_n = a^n\sin\left(ax+b+\frac{n\pi}{2}\right)$$

Ans.

* If $y = (ax+b)^m$ then check

$$y_n = \begin{cases} \dfrac{m!\,a^n(ax+b)^{m-n}}{(m-n)!} & ; m>0;\ m>n \\ 0 & ; m>0;\ m<n \\ n!\,a^n & ; m=n \\ \dfrac{(-1)^n n!\,a^n}{(ax+b)^{n+1}} & ; m=-1 \end{cases}$$

Example: Evaluate $\frac{d^n}{dx^n}(x^2 e^{ax})$ using Leibnitz theorem.

Ans:

let $u = e^{ax}$; $v = x^2$

$u_1 = \frac{du}{dx} = a e^{ax}$	$v_1 = \frac{dv}{dx} = 2x$
$u_2 = a^2 e^{ax}$	$v_2 = 2 - 0 = 2$
$\vdots$	$v_3 = 0$
$u_n = a^n e^{ax}$	

A.T. Leibnitz rule,

$$(uv)_n = \frac{d^n}{dx^n}(uv) = u_n v + {}^nc_1 u_{n-1} v_1 + {}^nc_2 u_{n-2} v_2 + {}^nc_3 u_{n-3} v_3 + 0$$

$$= a^n e^{ax} \cdot x^2 + \frac{n!}{1!(n-1)!} a^{n-1} e^{ax} \cdot 2x + \frac{n!}{2!(n-2)!} a^{n-2} \cdot e^{ax} \cdot 2 + \frac{n!}{3!(n-3)!} a^{n-3} e^{ax} \cdot 0$$

$$= a^n x^2 e^{ax} + n a^{n-1} 2x e^{ax} + \frac{n(n-1)}{2} a^{n-2} e^{ax} \cdot 2 + 0$$

Ans

Ex. If $y = \log\left(x+\sqrt{1+x^2}\right)$,

prove that,

$$(1+x^2)\,y_{n+2} + (2n+1)\,x\,y_{n+1} + n^2 y_n = 0.$$

<u>Ans.</u> Given

$$y = \log\left(x+\sqrt{1+x^2}\right)$$

$$\Rightarrow y_1 = \frac{dy}{dx} = \frac{d}{dx}\log\left(x+\sqrt{1+x^2}\right)$$

$$\Rightarrow y_1 = \frac{1+\frac{2x}{2\sqrt{1+x^2}}}{x+\sqrt{1+x^2}}$$

$$\Rightarrow y_1 = \frac{\frac{2\sqrt{1+x^2}+2x}{2\sqrt{1+x^2}}}{x+\sqrt{1+x^2}}$$

$$\Rightarrow y_1 = \frac{1}{\sqrt{1+x^2}}$$

$$\Rightarrow \sqrt{1+x^2}\,y_1 = 1$$

$$\Rightarrow (1+x^2)\,y_1^2 = 1$$

$$\Rightarrow \frac{d}{dx}\left\{(1+x^2)\,y_1^2\right\} = \frac{d}{dx}(1)$$

$$\Rightarrow (1+x^2)\,2y_1 y_2 + 2x\,y_1^2 = 0$$

$$\Rightarrow (1+x^2)\,y_2 + x\,y_1 = 0 \quad \rightarrow (1)$$

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using Leibnitz's theorem in ①

$$\Rightarrow \frac{d^n}{dx^n}\left\{(1+x^2)y_2\right\} + \frac{d^n}{dx^n}\left\{x y_1\right\} = 0$$

$$\Rightarrow y_{n+2}(1+x^2) + {}^nc_1\, y_{n+1}\cdot 2x + {}^nc_2\, y_n \cdot 2$$

$$+ {}^nc_3\, y_{n-1}\cdot 0 + y_{n+1}\, x + {}^nc_1\, y_n \cdot 1$$

$$+0 = 0$$

$$\Rightarrow (1+x^2)\, y_{n+2} + 2n\, x\, y_{n+1} + \frac{n(n-1)}{2} y_n \cdot 2$$

$$+ y_{n+1}\cdot x + n y_n = 0$$

$$\Rightarrow (1+x^2)\, y_{n+2} + (2n+1)\, x\, y_{n+1} + n^2 y_n = 0$$

(proved)

H.W. If $y = x^{n-1} e^{1/x}$; prove that

$$y_n = (-1)^n \frac{e^{1/x}}{x^{n+1}}.$$

practice sheet:

Exercise 2.4 — (5-20), 21, 24

Exercise - 2.6 — Example - 5, 6

Ex - 3, 4, 7-40, 43-54

Exercise - 2.3 - 29, 30, 31, 32, 41-48

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